

Abstract

Music education is frequently promoted as a way to improve children's cognitive development and academic performance. However, prior research on the topic has produced mixed findings, often due to differences in methods used and the underlying role of socioeconomic status. This project examines the relationship between music education and academic performance using data from the Panel Study of Income Dynamics Child Development Supplement (PSID-CDS) for 2014, 2019, and 2021. After data cleaning and normalization, 1,173 students remained in the sample. Statistical methods included descriptive analysis, t-tests, correlations, regression, and predictive modeling.

Results show that direct measures of music participation - taking lessons or general music involvement - were weak and inconsistent predictors of GPA. In contrast, having a musical instrument at home was the most stable and statistically significant predictor across all three years. The combined regression model explained 5.9% of the variance in GPA ($R^2 = 0.059$), suggesting that academic performance is influenced by factors beyond music variables. Predictive models also demonstrated limited explanatory power, reinforcing the complexity of academic performance.

The findings suggest that music education contributes to student success indirectly, through broader factors like household and environmental influences, rather than as a strong independent cause of higher academic performance. This project supports previous findings that music education is a nuanced component of child development, rather than a standalone determinant of academic performance and outcomes.